

Precise Attribution of Model Interventions: Attribution Sets, Mechanism Reach and Path Defects

A companion to *The Three-Level Posterior Tower*

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Abstract

A language model is trained on a corpus, extended with a second corpus, edited directly in its weights, and tuned with reinforcement learning from human feedback. Each step changes some outputs and leaves others alone, and new outputs appear. We ask three questions about such a history: what explains each observed change, how precisely the data determine that explanation, and whether the answer depends on the order of the steps. We answer them with three notions. An *attribution set* collects every decomposition of an observed output distribution into declared sources and an unmodeled remainder; it is a polytope, and its emptiness, its range of unmodeled weight and its certificates are computed by linear programming. *Precision* is the width of the attribution set; it is monotone along a ladder from exact sources to sources known by their supports to a positive budget for unmodeled sources, and any slack in that budget forces imprecision. *Mechanism reach* declares what an intervention type may change: we prove that the reach set of KL-regularized reinforcement learning with a bounded reward range is a polytope, so that attributing a change to it is a linear program with an exact certificate, whereas a fixed reward with unknown temperature gives a curve, and an unconstrained edit can explain anything. Finally, interventions are compared quantitatively: comparisons compose by the triangle inequality and by a Lipschitz rule, with exact Lipschitz constants for mixing and bounds for tilts, and a weight-level intervention may have no Lipschitz constant on observed outputs at all. Order and loop defects measure path dependence, and coarse evaluations certify fine differences. A synthetic four-stage experiment produces a full provenance report. Every theorem is checked in exact arithmetic or with exactly verified linear programming certificates.

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*Verification scripts: `provenance.py`, `check_attribution.py`, `provenance_demo.py`; see [appendix A](#).

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1 Introduction

Consider the life of a model. Corpus C_1 produces model M_1 . Adding corpus C_2 produces M_2 , which changes some outputs, leaves others unchanged, and makes outcomes probable that were not before. An engineer then edits weights directly, producing M_3 , and reinforcement learning from human feedback (RLHF) produces M_4 . For such a history we want answers to three questions.

- (Q1) *Attribution*. Which declared source or intervention explains each observed change, and what is left unexplained?
- (Q2) *Precision*. How tightly do the observations determine that explanation?
- (Q3) *Path dependence*. Does the result depend on the order of interventions, and does a sequence that should cancel actually cancel?

Architecture. The companion paper [3] develops the *local* structure: an ambient probability monad D , and over each outcome base B a context comonad, a mixed distributive law, admissibility data and an obstruction tower. Its §5 establishes identifiability and contamination results for a single output distribution. The present paper builds the *transport* half of the glue between local structures: interventions as maps between model states, and quantitative comparisons between interventions. The other half of the glue, assembling several contexts into a global state, is left to future work (section 8).

Contributions.

1. Attribution sets for declared sources known exactly or by their supports, with an unmodeled budget; their structure, emptiness criterion and certificates (theorem 3.2).
2. Precision as width, a monotone precision ladder separating structural, informational and unmodeled imprecision (theorem 3.5), and the fact that budget slack forces imprecision (proposition 3.6).
3. Mechanism classes and a change-decomposition theorem with certificates (theorem 4.8); the reach set of KL-regularized RLHF with bounded rewards is a polytope (theorem 4.4), while a fixed reward with unknown temperature is not convex (proposition 4.6).
4. Quantitative comparison of interventions with vertical and horizontal composition rules and an accumulation bound (theorem 5.5); exact or bounded Lipschitz constants for mixing, tilts and coarsening, and the absence of any observable Lipschitz constant for some weight-level interventions (propositions 5.3 and 5.4); order and loop defects; monotonicity under coarsening (proposition 5.9).
5. A synthetic four-stage experiment with a complete provenance report (section 6).

Several ingredients are known in other fields; section 7 states precisely which, and what is new here.

What is not claimed. We do not extract hypotheses from weights: declared sources must come from reference models, checkpoints or ablations. All guarantees concern a finite probe set. Attribution relative to a declared mechanism is not identification of a cause; causal statements require running the compared paths. We treat output distributions as known; finite-sample precision is an open problem (section 8).

2 Model states and observation

Definition 2.1 (Observable states). Fix a finite set X of *probes* (prompts or evaluation contexts) and, for each $x \in X$, a finite set B_x of outcomes, which may be coarse classes of completions. The *observable state space* is $\mathcal{P} = \prod_{x \in X} \mathcal{D}(B_x)$, with

$$d(p, q) = \max_{x \in X} \text{TV}(p_x, q_x), \quad \text{TV}(p_x, q_x) = \frac{1}{2} \sum_b |p_x(b) - q_x(b)|.$$

Definition 2.2 (Model states and interventions). A set $\mathcal{W}$ of *model states* (for instance weight vectors) carries an *observation map* $\Phi: \mathcal{W} \rightarrow \mathcal{P}$ sending a model to its output distributions on the probes. An *intervention* is a map $T: \mathcal{W} \rightarrow \mathcal{W}$; its *observable effect* on M is the pair $(\Phi M, \Phi TM)$.

Remark 2.3 (Observation is not faithful). Different models can have the same outputs on every probe, and the same intervention can then produce different outputs. This is why statements about interventions on weights cannot in general be reduced to statements about observable states (proposition 5.4).

Definition 2.4 (New outcomes). A softmax output assigns positive probability to every outcome, so “a new outcome appears” is made precise by a threshold $\varepsilon > 0$: outcome b *appears at level ε* at probe x under $M \rightarrow M'$ if $(\Phi M)_x(b) \leq \varepsilon < (\Phi M')_x(b)$.

From section 3 to section 4 we fix a probe, write B for its finite outcome set and $p \in \mathcal{D}(B)$ for an observed output distribution. All statements apply probe by probe.

3 Attribution sets and precision

Definition 3.1 (Declared sources and attribution sets). A *declared source* is either *exact*, a distribution $A_i \in \mathcal{D}(B)$, or *known by its support*, a set $S_i \subseteq B$. An *unmodeled budget* is a number $\bar{\mu} \in [0, 1]$. A *decomposition* of p is a family of nonnegative measures $s_1, \dots, s_k$ on B , the *contributions*, such that

- (a) $s_i = \lambda_i A_i$ for some $\lambda_i \geq 0$ if source i is exact, and $\text{supp } s_i \subseteq S_i$ if source i is known by its support, in which case $\lambda_i = \text{mass}(s_i)$;
- (b) $\sum_i s_i \leq p$ pointwise, and $\mu := 1 - \sum_i \lambda_i \leq \bar{\mu}$.

The *unmodeled part* is $p - \sum_i s_i = \mu U$ with $U \in \mathcal{D}(B)$ when $\mu > 0$. The *attribution set* $W_{\bar{\mu}}(p)$ is the set of all decompositions. The *minimal unmodeled weight* $\mu_{\min}(p)$ is the least μ over decompositions with $\bar{\mu} = 1$.

The model is Huber’s contamination model [13] with declared sources. A decomposition determines U , so the attribution set records exactly the explanations compatible with p .

Theorem 3.2 (Structure of attribution sets).

- (i) $W_{\bar{\mu}}(p)$ is a polytope, cut out by linear constraints on the contributions.
- (ii) $W_{\bar{\mu}}(p) \neq \emptyset$ if and only if $\mu_{\min}(p) \leq \bar{\mu}$.
- (iii) If $W_{\bar{\mu}}(p) \neq \emptyset$, the unmodeled weights it contains form the interval $[\mu_{\min}(p), \bar{\mu}]$.
- (iv) (Certificate) Suppose $y \in \mathbb{R}_{\geq 0}^B$ satisfies $\langle y, a \rangle \geq 1$ for every admissible version a of every source, that is $\langle y, A_i \rangle \geq 1$ for exact sources and $y_b \geq 1$ for $b \in S_i$ for support-known sources. Then every decomposition has $\mu \geq 1 - \langle y, p \rangle$, and the best such bound equals $\mu_{\min}(p)$.

Proof. (i) Constraints (a) and (b) are linear in the variables λ_i (exact sources) and s_i (support-known sources), with $s_i \geq 0$, and the feasible set is bounded since $\sum_i s_i \leq p$. (ii) and (iii) If a decomposition has weight vector λ , then so does $t\lambda$ (with scaled contributions) for every $t \in [0, 1]$, which raises μ continuously up to 1. Hence the attainable μ form $[\mu_{\min}(p), 1]$, and the budget cuts this to $[\mu_{\min}(p), \bar{\mu}]$. (iv) Each contribution is a nonnegative multiple $\lambda_i a$ of an admissible version a of its source, so $\langle y, \sum_i s_i \rangle \geq \sum_i \lambda_i = 1 - \mu$, while $\langle y, \sum_i s_i \rangle \leq \langle y, p \rangle$. Optimality of the best bound is linear programming duality [27]: $1 - \mu_{\min}(p)$ is the value of $\max\{\sum_i \lambda_i\}$ over (a), (b), and its dual is $\min\{\langle y, p \rangle\}$ under the stated constraints on y . $\square$

Remark 3.3 (Mixture proportion estimation). With a single exact source A , $\mu_{\min}(p) = 1 - \kappa^*(p|A)$, where $\kappa^*(p|A) = \min_{b \in \text{supp } A} p(b)/A(b)$ is the maximal proportion of A in p studied in mixture proportion estimation [4, 28]. A positive budget $\bar{\mu}$ plays the role of the bound on the error probability in contaminated sampling [12].

Definition 3.4 (Precision). For a nonempty attribution set, the *width* of source i is $w_i = \max \lambda_i - \min \lambda_i$ over $W_{\bar{\mu}}(p)$, and the width of the unmodeled weight is $w_{\mu} = \max \mu - \min \mu$. For an outcome b with $p(b) > 0$, the *observation-level attribution* $\pi_i(b) = s_i(b)/p(b)$ is the probability that an observation of b came from source i ; its width is defined likewise. The attribution is *exact* if all widths vanish. All widths are values of linear programs over $W_{\bar{\mu}}(p)$.

Theorem 3.5 (Monotone precision ladder). *Enlarging the admissible versions of any source, or raising the budget, can only enlarge the attribution set; hence all widths are nondecreasing. In particular, writing $\mu_{\min}^{\text{ex}}$ for the minimal unmodeled weight with exact sources, the attribution sets*

$$W^{(1)} = W_{\mu_{\min}^{\text{ex}}}^{\text{exact}}(p) \subseteq W^{(2)} = W_{\mu_{\min}^{\text{ex}}}^{\text{support}}(p) \subseteq W^{(3)} = W_{\bar{\mu}}^{\text{support}}(p) \quad (\bar{\mu} \geq \mu_{\min}^{\text{ex}})$$

are nonempty and nested, and each width satisfies $w^{(1)} \leq w^{(2)} \leq w^{(3)}$.

Proof. Every decomposition with exact sources is one with the corresponding supports, since $\text{supp}(\lambda_i A_i) \subseteq \text{supp } A_i$; and every decomposition within budget $\bar{\mu}$ is one within any larger budget. The width of a coordinate over a larger set is at least its width over a smaller set. $W^{(1)}$ is nonempty by [theorem 3.2\(ii\)](#). $\square$

We call the increments $w^{(1)}$, $w^{(2)} - w^{(1)}$ and $w^{(3)} - w^{(2)}$ the *structural*, *informational* and *unmodeled* imprecision. They are nonnegative by [theorem 3.5](#); we do not claim that they combine additively across sources.

Proposition 3.6 (Budget slack forces imprecision). *If $\mu_{\min}(p) < \bar{\mu}$ and $\mu_{\min}(p) < 1$, then $w_{\mu} = \bar{\mu} - \mu_{\min}(p) > 0$, and every source whose maximal weight is positive has positive width.*

Proof. The first claim is [theorem 3.2\(iii\)](#). Take a decomposition λ with $\mu = \mu_{\min}(p)$. For $t = (1 - \bar{\mu})/(1 - \mu_{\min}(p)) < 1$, the scaled family $t\lambda$ is again a decomposition, since its unmodeled weight is $\bar{\mu}$. If $\lambda_i > 0$, the two decompositions give different values of the i -th weight. If $\lambda_i = 0$, this value differs from the positive maximal weight. Either way $w_i > 0$. $\square$

So precision and robustness trade off: a budget for unmodeled sources protects against rejecting the declared model, and every unit of slack in that budget is paid for in width.

Proposition 3.7 (When attribution is exact). *With exact sources and budget $\mu_{\min}(p) = 0$, the attribution is exact for every p in $\text{conv}(A_1, \dots, A_k)$ if and only if the sources are affinely independent. For any sources, if the exclusive region $R_i = \{b : A_i(b) > 0, A_j(b) = 0 (j \neq i)\}$ has $A_i(R_i) > 0$, then $\lambda_i = p(R_i)/A_i(R_i)$ in every decomposition, and $\pi_i(b) = \lambda_i A_i(b)/p(b)$.*

Proof. This is [3, Prop. 5.7]. □

4 Mechanism classes and attribution of change

Definition 4.1 (Mechanism class). A *mechanism class* for an intervention type assigns to each $p \in \mathcal{D}(B)$ a set $\text{Reach}(p) \subseteq \mathcal{D}(B)$ of output distributions the intervention may produce from p . A change $p \rightarrow p'$ is *reachable* if $p' \in \text{Reach}(p)$.

Example 4.2 (Three polyhedral classes).

- (a) *Source expansion* (adding a corpus): $\text{Reach}(p) = \text{conv}(\mathcal{A} \cup \mathcal{N})$, where $\mathcal{A}$ are the old sources and $\mathcal{N}$ the new ones.
- (b) *Locality claim* for an edit with declared scope $X_0 \subseteq X$ and tolerance τ : at a probe outside X_0 , $\text{Reach}(p) = \{q : \text{TV}(q, p) \leq \tau\}$.
- (c) *Unconstrained edit*: $\text{Reach}(p) = \mathcal{D}(B)$.

Proposition 4.3 (KL-regularized reinforcement learning is an exponential tilt). *For a reward $r : B \rightarrow \mathbb{R}$ and $\beta > 0$, the maximizer of $\mathbb{E}_q[r] - \beta \text{KL}(q \| p)$ over $q \in \mathcal{D}(B)$ is*

$$T_r(p)(b) = \frac{p(b) e^{r(b)/\beta}}{\sum_c p(c) e^{r(c)/\beta}}.$$

Proof. For $q \ll p$, $\mathbb{E}_q[r] - \beta \text{KL}(q \| p) = \beta \log Z - \beta \text{KL}(q \| T_r(p))$ with $Z = \sum_c p(c) e^{r(c)/\beta}$, which is maximal exactly at $q = T_r(p)$; and $q \not\ll p$ gives $-\infty$. This is the standard Gibbs variational principle, used for RLHF in [17, 24]. □

Writing $w(b) = e^{(r(b) - r_{\min})/\beta}$, a reward with values in $[r_{\min}, r_{\max}]$ gives weights in $[1, K]$ with $K = e^{(r_{\max} - r_{\min})/\beta}$, and $T_r(p)$ depends only on w . The tilt is a reweighting by a likelihood, the operation called conditioning in [3, §4]; it is not conditioning on an observed event. It preserves support exactly, $\text{supp } T_r(p) = \text{supp } p$, but it can raise a small probability by a factor up to K .

Theorem 4.4 (The tilt reach set is a polytope). *Let $\text{Reach}_K(p) = \{T_r(p) : r(B) \subseteq [r_{\min}, r_{\max}]\}$ with K as above. Then*

$$\text{Reach}_K(p) = \{q \in \mathcal{D}(B) : \text{supp } q = \text{supp } p, \quad q_b p_c \leq K q_c p_b \text{ for all } b, c \in \text{supp } p\},$$

a convex polytope. Equivalently $q \in \text{Reach}_K(p)$ iff $\text{supp } q = \text{supp } p$ and

$$K^*(p, q) = \frac{\max_{b \in \text{supp } p} q_b / p_b}{\min_{b \in \text{supp } p} q_b / p_b} \leq K,$$

so $\beta \log K^(p, q)$ is the least reward range that explains the change.*

Proof. If $q = T_r(p)$, then $q_b/p_b = w_b/Z$ for $b \in \text{supp } p$, so the ratios lie within a factor K of each other, which is the stated inequality. Conversely, if q satisfies the inequalities and has support $\text{supp } p$, put $w_b = (q_b/p_b)/\min_c(q_c/p_c) \in [1, K]$ for $b \in \text{supp } p$ and $w_b = 1$ elsewhere; then $T_w(p) = q$. The inequalities are linear in q for fixed p ; together with $\sum_b q_b = 1$ and $q_b = 0$ off $\text{supp } p$ they define a polytope. A point satisfying them cannot vanish anywhere on $\text{supp } p$: if $q_b = 0$ then $q_c p_b \leq K q_b p_c = 0$ for all c , contradicting $\sum q = 1$. So the constraint $\text{supp } q = \text{supp } p$ is implied, and the set is closed. $\square$

Remark 4.5 (Density ratio classes). The set in [theorem 4.4](#) is the interval of measures $\{q : p \leq cq \leq Kp \text{ for some } c > 0\}$ of DeRobertis and Hartigan [8], known in robust Bayesian analysis as a density ratio class. [Theorem 4.4](#) therefore identifies the reach of KL-regularized RLHF with bounded rewards as a classical object; what it adds is that identification, and the reading of $\beta \log K^*$ as the least reward range that explains an observed change.

Proposition 4.6 (Fixed reward, unknown temperature). *For a fixed reward r , the set $\{T_{r/\beta'}(p) : \beta' > 0\}$ is a curve, and in general not convex: for p uniform on three outcomes and $r = (0, 1, 2)$, the midpoint of the points with $1/\beta' = 1$ and $1/\beta' = 3$ is not on the curve.*

Proof. On the curve, $\log(q_b/p_b) - \log(q_0/p_0)$ is proportional to $r_b - r_0$, so $\log(q_2/q_0) = 2 \log(q_1/q_0)$ for this p and r ; the midpoint violates this identity, as checked in [appendix A](#). $\square$

So the geometry of a reach set depends on what is assumed about the mechanism: a bounded reward range gives a polytope, a fixed reward gives a curve, and no assumption gives everything.

Definition 4.7 (Residual of a change). Let $\text{cone Reach}(p) = \{tq : t \geq 0, q \in \text{Reach}(p)\}$. An *explained part* of a change $p \rightarrow p'$ is a measure $s \in \text{cone Reach}(p)$ with $s \leq p'$; the remainder $p' - s$ is the *residual*. The *residual weight* of the change is

$$\mu_{\min}^T(p, p') = 1 - \max\{\text{mass}(s) : s \in \text{cone Reach}(p), s \leq p'\}.$$

Theorem 4.8 (Change decomposition). *Let $\text{Reach}(p)$ be closed and convex.*

- (i) *The maximum defining $\mu_{\min}^T(p, p')$ is attained, and $\mu_{\min}^T(p, p') = 0$ if and only if $p' \in \text{Reach}(p)$.*
- (ii) *(Certificate) If $y \geq 0$ satisfies $\langle y, q \rangle \geq 1$ for all $q \in \text{Reach}(p)$, then $\mu_{\min}^T(p, p') \geq 1 - \langle y, p' \rangle$. If $\text{cone Reach}(p) = \{s \geq 0 : Gs \leq 0\}$ is polyhedral, the best such bound is attained by a pair $y \geq 0, \nu \geq 0$ with $y + G^\top \nu \geq \mathbf{1}$, and it equals $\mu_{\min}^T(p, p')$.*
- (iii) *(Precision of change attribution) The optimal explained parts form a polytope when the cone is polyhedral, and the range of each $s(b)$ over it is given by linear programs.*

Proof. (i) $\text{Reach}(p) \subseteq D(B)$ is compact and avoids 0, so its cone is closed; intersected with $\{0 \leq s \leq p'\}$ it is compact, and mass is continuous. If the maximum is 1, then $s \leq p'$ and $\text{mass}(s) = \text{mass}(p')$ force $s = p'$, so $p' \in \text{cone Reach}(p)$ with mass one, that is, $p' \in \text{Reach}(p)$. (ii) For $s = tq$ with $q \in \text{Reach}(p)$, $\text{mass}(s) = t \leq t\langle y, q \rangle = \langle y, s \rangle \leq \langle y, p' \rangle$. In the polyhedral case, $\max\{\mathbf{1}^\top s : Gs \leq 0, s \leq p', s \geq 0\}$ has dual $\min\{\langle y, p' \rangle : y + G^\top \nu \geq \mathbf{1}, y, \nu \geq 0\}$, and strong duality holds [27]; every dual-feasible pair satisfies $\langle y, s \rangle \geq \mathbf{1}^\top s - \langle \nu, Gs \rangle \geq \mathbf{1}^\top s$ on the cone. (iii) The optimal set is a face of a polytope. $\square$

Remark 4.9 (What residuals mean). A positive residual weight says that at least this much of the observed change is not produced by any intervention of the declared type. It does not say what produced it. For the tilt class the cone is $\{s \geq 0 : s_b p_c \leq K s_c p_b, s = 0 \text{ off } \text{supp } p\}$, so attributing a change to RLHF is a linear program with an exact certificate. For an unconstrained edit the residual is always zero: an edit can be tested only through a claim about it, such as locality.

5 Comparing interventions: composition and path defects

Definition 5.1 (Quantitative comparison). For interventions F, G and a set $\mathcal{S} \subseteq \mathcal{W}$ of model states,

$$d_{\mathcal{S}}(F, G) = \sup_{M \in \mathcal{S}} d(\Phi FM, \Phi GM).$$

This is a pseudometric on interventions. We write $F \approx_{\varepsilon} G$ on $\mathcal{S}$ if $d_{\mathcal{S}}(F, G) \leq \varepsilon$, and call this a *quantitative comparison*.

A quantitative comparison records how far two interventions are apart; it is not by itself a 2-cell in a 2-category, and we do not call it one. [Corollary 5.7](#) identifies the structure in which it is the hom-metric of an enriched category.

Definition 5.2 (Lipschitz on observables). An intervention G is *L -Lipschitz on observables* over $\mathcal{T} \subseteq \mathcal{W}$ if $d(\Phi GM, \Phi GM') \leq L d(\Phi M, \Phi M')$ for all $M, M' \in \mathcal{T}$. An *output-level mechanism* is a map $T: \mathcal{P} \rightarrow \mathcal{P}$ acting on observable states; for it the Lipschitz constant is the usual one for d .

Proposition 5.3 (Lipschitz constants of output-level mechanisms).

- (i) *Mixing with a fixed target, $p \mapsto (1 - a)p + at$, is exactly $(1 - a)$ -Lipschitz.*
- (ii) *A tilt with weights in $[1, K]$ is $\frac{3K-1}{2}$ -Lipschitz, and no constant below K works in general.*
- (iii) *Coarsening along any map of outcomes is 1-Lipschitz.*

Proof. (i) The difference of images is $(1 - a)(p - q)$. (iii) Data processing for total variation. (ii) Let $Z = \langle w, p \rangle \geq 1$ and $Z' = \langle w, q \rangle$. Then $T(p) - T(q) = w \odot (p - q)/Z + w \odot q(1/Z - 1/Z')$, so $\|T(p) - T(q)\|_1 \leq (K\|p - q\|_1 + |Z - Z'|)/Z \leq K\|p - q\|_1 + |Z - Z'|$, using $\|w \odot q\|_1 = Z'$ and $Z \geq 1$. Since $p - q$ has total mass zero, $|Z - Z'| = |\langle w - \frac{K+1}{2}\mathbf{1}, p - q \rangle| \leq \frac{K-1}{2}\|p - q\|_1$. For the lower bound take $p = (\epsilon, 1 - \epsilon)$ and $w = (K, 1)$: the first coordinate of $T(p)$ is $K\epsilon/(1 + (K - 1)\epsilon)$, whose derivative at $\epsilon = 0$ is K . $\square$

So tilts can amplify differences, which matters for composition.

Proposition 5.4 (No observable Lipschitz constant). *If $\Phi M = \Phi M'$ but $\Phi GM \neq \Phi GM'$, then G is not L -Lipschitz on observables over $\{M, M'\}$ for any L .*

Proof. The left side of the Lipschitz inequality is positive and the right side is zero. $\square$

A weight edit that depends on parts of the model not visible on the probes is of this kind. For such interventions the composition bound below must use a metric on $\mathcal{W}$, and conclusions at the probe level are conditional on that metric.

Theorem 5.5 (Composition of quantitative comparisons).

- (i) (Vertical) $d_{\mathcal{S}}(F, H) \leq d_{\mathcal{S}}(F, G) + d_{\mathcal{S}}(G, H)$.
- (ii) (Horizontal) If G' is L -Lipschitz on observables over $F(\mathcal{S}) \cup F'(\mathcal{S})$, then

$$d_{\mathcal{S}}(GF, G'F') \leq d_{F(\mathcal{S})}(G, G') + L d_{\mathcal{S}}(F, F').$$

(iii) (Accumulation) For paths $F_n \cdots F_1$ and $G_n \cdots G_1$, let $\mathcal{S}_0 = \mathcal{S}$ and $\mathcal{S}_j = F_j(\mathcal{S}_{j-1})$, put $\varepsilon_j = d_{\mathcal{S}_{j-1}}(F_j, G_j)$, and suppose each G_j is L_j -Lipschitz on observables over the states reached by both paths. Then

$$\delta_j = d_{\mathcal{S}}(F_j \cdots F_1, G_j \cdots G_1) \quad \text{satisfies} \quad \delta_j \leq \varepsilon_j + L_j \delta_{j-1},$$

and hence

$$\delta_n \leq \sum_{j=1}^n \varepsilon_j \prod_{i=j+1}^n L_i.$$

Proof. (i) is the triangle inequality for d . (ii) For $M \in \mathcal{S}$,

$$d(\Phi GFM, \Phi G'F'M) \leq d(\Phi GFM, \Phi G'FM) + d(\Phi G'FM, \Phi G'F'M);$$

the first term is at most $d_{F(\mathcal{S})}(G, G')$ and the second at most $L d(\Phi FM, \Phi F'M)$. (iii) Apply (ii) with $F = F_j$, $G = G_j$ to the states $F_{j-1} \cdots F_1 M$ and $G_{j-1} \cdots G_1 M$, and unroll the recursion. $\square$

Remark 5.6. Composition rules of this kind are standard for behavioral pseudometrics of probabilistic systems [9], in quantitative algebraic reasoning [20] and in sensitivity type systems [25]. [Theorem 5.5](#) is an instance; what is specific here is the observation map Φ , which is not faithful, and the constants for tilts.

Corollary 5.7 (Enriched structure). *Output-level mechanisms on $\mathcal{P}$ with Lipschitz constant at most 1, with $d = d_{\mathcal{P}}$, form a category enriched in pseudometric spaces in Lawvere's sense [18]: $d(\text{id}, \text{id}) = 0$ and $d(GF, G'F') \leq d(G, G') + d(F, F')$. Mixings and coarsenings belong to it; tilts do not, since their Lipschitz constant can exceed 1, so in general the structure is the Lipschitz-weighted one of [theorem 5.5](#).*

Definition 5.8 (Order and loop defects). For interventions E and R , the *order defect* on $\mathcal{S}$ is $d_{\mathcal{S}}(RE, ER)$. For a sequence of interventions $T_1, \dots, T_n$, the *loop defect* is $d_{\mathcal{S}}(T_n \cdots T_1, \text{id})$.

A positive order defect certifies that the two orders produce different observable models on the probes; it becomes a causal statement once both orders have actually been run. We call the loop defect a defect rather than a holonomy, since no connection has been defined.

Proposition 5.9 (Coarse evaluation). *For any coarsening g of outcomes, $\text{TV}(g_*p, g_*q) \leq \text{TV}(p, q)$. Hence every order, loop or comparison defect measured on a coarse evaluation is a lower bound for the fine one, and a positive coarse defect certifies a positive fine defect.*

6 A four-stage provenance experiment

We run a synthetic history through all four stages and report what the framework certifies. The experiment is in `provenance_demo.py`; all distributions are exact rationals.

Setup. There are three probes x_1, x_2, x_3 and six outcomes $y_1, \dots, y_6$. Corpus C_1 has two sources A_1, A_2 , and corpus C_2 has one source N , which is the only one giving mass to y_6 :

probe	A_1	A_2	N
x_1	(.4, .3, .2, .1, 0, 0)	(0, .1, .2, .3, .4, 0)	(0, 0, 0, .2, .3, .5)
x_2	(.1, .1, .4, .4, 0, 0)	(.3, .3, .1, .1, .2, 0)	(0, 0, 0, 0, .5, .5)
x_3	(.5, .5, 0, 0, 0, 0)	(0, 0, .5, .5, 0, 0)	(0, 0, 0, 0, .5, .5)

M_1 mixes A_1, A_2 with weights $(0.6, 0.4)$ at every probe. M_2 uses weights $(0.45, 0.25, 0.30)$ at x_1 , $(0.5, 0.3, 0.2)$ at x_2 , and leaves x_3 unchanged. The edit producing M_3 declares scope $\{x_1\}$ and moves 20% of the mass at x_1 to y_3 , but also leaks 5% to y_1 at x_2 . RLHF producing M_4 is a tilt with weights $(1, 2, 4, 1, 3, 1)$, so $K = 4$; a *drifted* variant additionally multiplies y_6 by 10 at x_2 .

Stage 2: adding C_2 . With exact sources the attribution is exact at every probe: the weights (A_1, A_2, N) are recovered as $(0.45, 0.25, 0.30)$, $(0.5, 0.3, 0.2)$ and $(0.6, 0.4, 0)$, with widths zero. Outcome y_6 appears at level $\varepsilon = 0.01$ at x_1 and x_2 and not at x_3 , where $\text{TV}(M_1, M_2) = 0$. Since y_6 is exclusive to N , [proposition 3.7](#) gives $\lambda_N = p(y_6)/N(y_6)$, namely $3/10$ and $1/5$. An observation of y_5 at x_1 came from A_2 with probability $10/19$ and from N with probability $9/19$. The precision ladder at x_1 is:

stage	A_1	A_2	N	μ
exact sources, budget 0	0.45	0.25	0.30	0
supports only, budget 0	$[0.18, 0.66]$	$[0, 0.67]$	$[0.15, 0.52]$	0
supports only, budget 0.1	$[0.08, 0.66]$	$[0, 0.67]$	$[0.05, 0.52]$	$[0, 0.1]$

The structural imprecision is zero, the informational imprecision is large, and the budget lowers the lower ends further, as [theorem 3.5](#) and [proposition 3.6](#) predict. The lower bound 0.18 for A_1 is the mass of its exclusive outcome y_1 .

Stage 3: the edit. The locality test with $\tau = 0.01$ passes at x_3 ($\text{TV} = 0$) and fails at x_2 ($\text{TV}(M_2, M_3) = 0.043$), detecting the leak. Relative to the corpus sources A_1, A_2, N , the certified unmodeled weight is 0.20 at x_1 and 0.05 at x_2 , exactly the mass moved by the edit and by the leak, with certificates y satisfying $\min_i \langle y, A_i \rangle = 1$ and $\langle y, p \rangle = 0.80$, respectively 0.95. At x_1 the total-variation change is 0.172, smaller than the certified unexplained weight 0.20: the two quantities answer different questions.

Stage 4: RLHF. For the pure tilt, the implied ratio is $K^* = 4$ at every probe, so every change is reachable with the claimed reward range. For the drifted variant, $K^* = 10$ at x_2 , so the change is not reachable; the residual weight relative to the tilt class is 0.1934, with an exactly verified certificate. The other probes remain reachable, so the departure from the declared mechanism is localized.

Paths. At x_1 , with E the edit and R the tilt, the order defect is $\text{TV}(REp, ERp) = 51/518 \approx 0.0985$, so the order matters. Tilting by w and then by $1/w$ has loop defect zero. Comparing E with a weaker edit E' (moving 25% instead of 20%), $\text{TV}(REp, RE'p) = 0.0479$ is within the bound $\frac{3K-1}{2}\text{TV}(Ep, E'p) = 5.5 \times 0.043$, with observed amplification 1.11. On the coarse evaluation $\{y_1, y_2\}, \{y_3, y_4\}, \{y_5, y_6\}$ the order defect is 0.0858, smaller but still positive, so the coarse evaluation alone certifies that the order matters.

Summary. The report separates what changed, what the declared sources and mechanisms explain, how precisely, and where they fail:

step	behavioral change	attribution	certified residual
add C_2	$\text{TV} = 0.18$ at x_1, x_2 ; 0 at x_3	exact; $\lambda_N = 0.3, 0.2, 0$	none
edit	$\text{TV} = 0.172$ at x_1 ; leak 0.043 at x_2	locality fails at x_2	0.20 (x_1), 0.05 (x_2)
RLHF	tilt, $K^* = 4$	reachable at all probes	none
RLHF, drifted	$K^* = 10$ at x_2	not reachable at x_2	0.19 (x_2)

7 Relation to prior work

Most individual ingredients of this paper are known; the contribution is their combination into a calculus for model provenance whose reported object is a set of explanations together with its width and a certificate. We state the relation area by area and then summarize.

Mixture proportion estimation and contaminated data. Mixture proportion estimation studies $F = (1 - \kappa)G + \kappa H$ with H known and G unknown; the maximal proportion $\kappa^*(F|H) = \text{ess inf } f/h$ is identified, and κ itself is identified under irreducibility [4, 28], with extensions beyond irreducibility [31]. Several known components plus one unknown appear in class-proportion estimation with anomaly rejection [26], in open-set label shift, where identifiability needs positivity or separability conditions [11], and in mutual contamination models [5]. In econometrics, Horowitz and Manski derive sharp bounds under contaminated sampling with a known bound on the error probability [12], within the program of partial identification [19]. Our minimal unmodeled weight is the finite-outcome complement of κ^* (remark 3.3), and our budget is the Horowitz–Manski error bound. These works estimate proportions from samples under identifying assumptions. We instead treat output distributions as known, characterize the *whole* set of decompositions whether or not it is identified, allow sources known exactly and sources known only by their supports in the same model, report sharp intervals for every weight and every observation-level attribution, and attach exact certificates.

Imprecise probability. Attribution sets are credal sets of decompositions in the sense of imprecise probability [30]; for sources known by their supports, the weight intervals are belief and plausibility of Dempster–Shafer theory [7, 29] ([3, Prop. 5.8]). The density ratio classes of robust Bayesian analysis [8] are exactly the reach sets of bounded-reward RLHF (remark 4.5). What is new here is the precision ladder that separates structural, informational and unmodeled imprecision (theorem 3.5), and the fact that slack in the unmodeled budget always costs precision (proposition 3.6).

RLHF. The Gibbs form of the optimum of KL-regularized reinforcement learning is known [17, 24]; we use it. The new point is a membership test for the resulting class of changes and a certified residual when a change falls outside it (theorems 4.4 and 4.8).

Training-data attribution. Influence functions [16], datamodels [15] and TRAK [23] estimate how individual training examples affect predictions, approximating counterfactual retraining. They answer a different question. We take sources as given, for instance reference models trained on each corpus, and certify what observed output distributions do and do not determine about them. The two are complementary: data-attribution estimates can be used to construct declared sources.

Model editing, diffing and composition. Locality or specificity of weight edits is measured as an evaluation metric [21], and edits are known to have ripple effects beyond their intended scope [6]. Here a locality claim is a mechanism class, and its violation comes with a certified residual (example 4.2, theorem 4.8). Crosscoder-based model diffing compares the internal features of a base and a fine-tuned model [22]; our comparisons are behavioral, at the probe outputs. Task arithmetic composes interventions as vectors in weight space [14]; the order defect measures whether composition commutes in the outputs.

Metric semantics. Categories enriched in generalized metric spaces go back to Lawvere [18]; behavioral pseudometrics for probabilistic systems [9], quantitative algebraic reasoning [20] and sensitivity type systems [25] all use composition rules with Lipschitz factors. Theorem 5.5 is an instance. What is specific here is that interventions act on model states and are compared only through a non-faithful observation map, which can rule out any Lipschitz constant (proposition 5.4), and the explicit constants for tilts (proposition 5.3).

Contextuality and comonadic simulation. The relation to contextuality is one of extension, not replacement. Abramsky and Brandenburger formulate an empirical model as compatible local probability data and characterize contextuality by the existence of a global section [2]. Abramsky, Barbosa, Karvonen and Mansfield then gave a comonadic treatment of simulations, that is, of morphisms between empirical models [1]. Morphisms are therefore not our novelty. Our local object is richer: for each base B it contains the ambient probability structure together with a base-dependent context comonad, a mixed distributive law, higher-order admissibility data and an obstruction tower [3]. The maps considered here are meant to transport these whole local structures, and the present paper equips such transport with quantitative comparison and path defects. The complementary global problem, whether compatible local structures assemble into a single coherent state, including under approximate compatibility, is left open.

Summary.

Result	Closest prior work	What is new here
Theorem 3.2	mixture proportion estimation; contaminated-sampling bounds	full polytope of decompositions for several exact or support-known sources; sharp weight and observation intervals; exact certificates
Theorem 3.5, proposition 3.6	credal sets; partial identification	monotone ladder of structural, informational and unmodeled imprecision; slack costs precision
Theorem 4.4	density ratio classes; Gibbs form of RLHF	RLHF reach is a density ratio class; implied reward range $\beta \log K^*$
Theorem 4.8	LP duality	residual of a change relative to a mechanism class, with certificate
Theorem 5.5, propositions 5.3 and 5.4	behavioral metrics; quantitative algebra; sensitivity typing	interventions seen through a non-faithful observation map; constants for tilts; no-Lipschitz obstruction
Section 6	—	an end-to-end provenance report with widths and certificates

We do not claim as new the linear-programming and duality machinery, the single-source formula of mixture proportion estimation, the Gibbs form of RLHF, density ratio classes, or the Lipschitz composition calculus.

8 Scope and open problems

Local structure and glue. In the architecture of [3], the probability monad D is ambient, and each base B carries local structure: a context comonad, a distributive law, admissibility data and an obstruction tower. This paper supplies the transport between local structures: interventions,

mechanism classes and quantitative comparison. Deterministic base change is already a structure-preserving map in [3, Prop. 9.13]; extending it to Markov kernels and to interventions observed through Φ is the next step.

Structural provenance. The obstruction tower of [3, §10] gives a second coordinate. For a claim c attached to a model, let $k(c)$ be its first failing level. Along an intervention, the pair $(k_{\text{before}}, k_{\text{after}})$ records which level of admissibility changed. Because failure is monotone up the tower [3, Thm. 10.5], a transition $k : 1 \rightarrow 2$ means that a level-one obstruction was removed while a level-two one remains; it does not mean that a level-two obstruction was created. We leave the systematic study of such transitions to future work.

Open problems.

1. *Finite-sample precision.* Output distributions are estimated from samples. Replacing p by a confidence region $\mathcal{C}_\delta$ turns each linear program into a robust program and each width into a statistical width.
2. *Weight-level metrics.* Choose a metric on $\mathcal{W}$ for which real interventions are Lipschitz, so that [theorem 5.5](#) applies beyond output-level mechanisms.
3. *Higher levels.* Lift comparisons to second- and third-order states with the Kantorovich metric [10], and relate them to the obstruction tower.
4. *Multi-context assembly.* Glue several probes or contexts into one global state, with approximate compatibility measured by quantitative comparisons.
5. *Empirical validation.* Run the provenance report on real checkpoints, with declared sources from reference models.

A Code and computational checks

The library `provenance.py` implements attribution sets, widths, residuals and certificates. The script `check_attribution.py` verifies the following. [theorem 3.2](#) (nonemptiness criterion, the interval $[\mu_{\min}, \bar{\mu}]$, and exactly verified dual certificates); [theorem 3.5](#) (monotone widths along the ladder and in the budget); [proposition 3.6](#); [theorem 4.4](#) in exact arithmetic (every tilt satisfies the inequalities, every point satisfying them is reconstructed exactly as a tilt, and mixtures of tilts are tilts); [proposition 4.6](#); [theorem 4.8](#) for the tilt class (exactly verified certificates (y, ν) , and zero residual exactly for reachable changes); [proposition 5.3](#) (the exact constant for mixing, the tilt bound, and an example with amplification 4.0 for $K = 4$); [proposition 5.4](#); [theorem 5.5\(i\)–\(iii\)](#) in exact arithmetic; and [proposition 5.9](#). All 18 checks pass. The script `provenance_demo.py` produces the report of [section 6](#). Linear programs are solved in floating point, and their certificates are rounded to rationals and verified exactly.

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